



NEWSLETTER

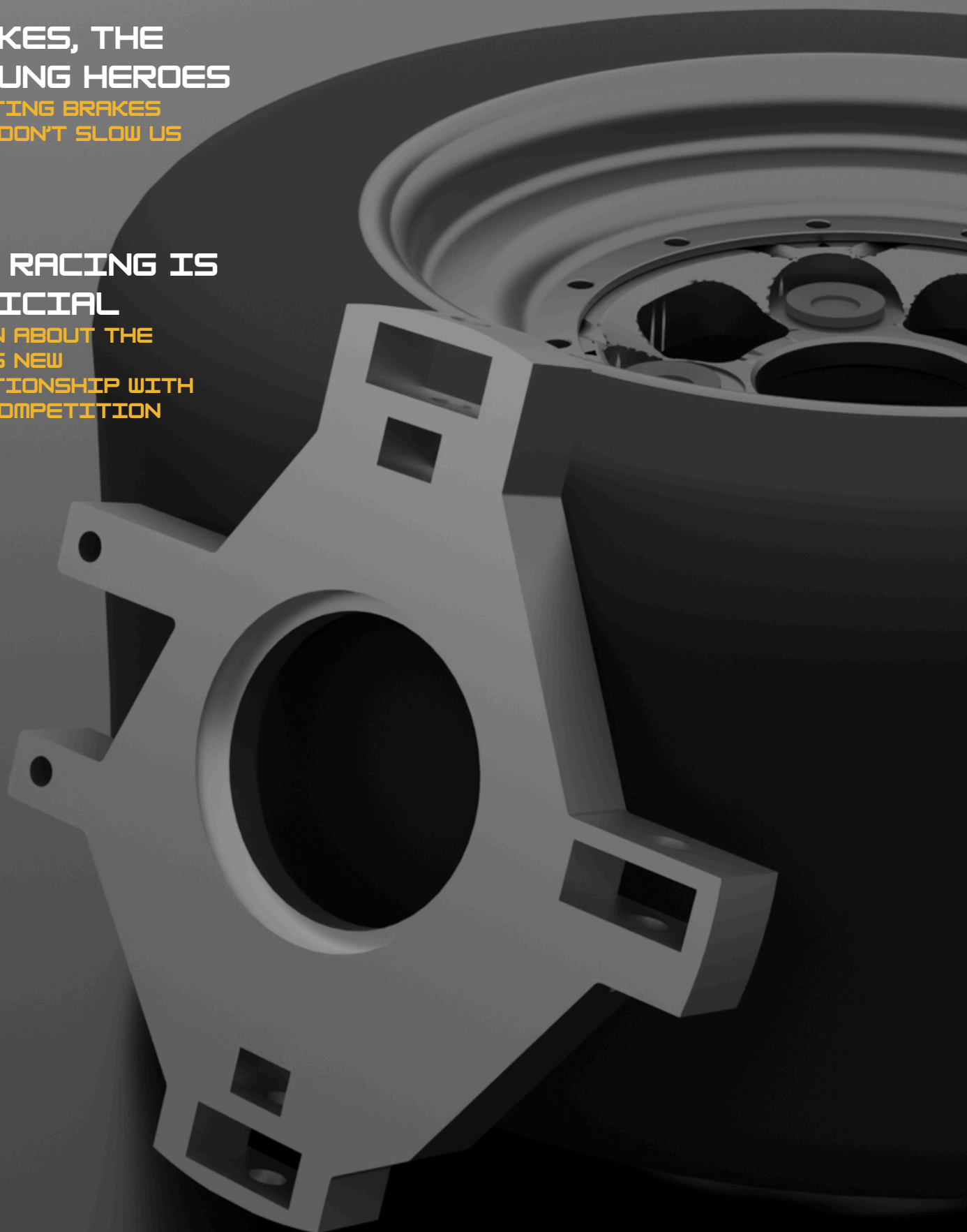
JANUARY 2024

BRAKES, THE UNSUNG HEROES

CREATING BRAKES
THAT DON'T SLOW US
DOWN.

AUS RACING IS OFFICIAL

LEARN ABOUT THE
TEAMS NEW
RELATIONSHIP WITH
THE COMPETITION



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WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

MAKE IT STOP



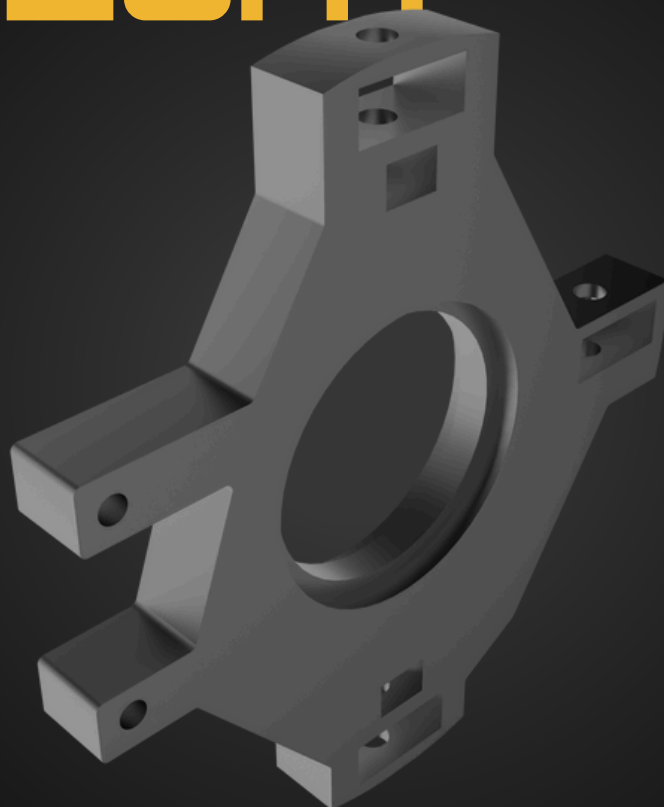
The mechanical brakes of the car are extremely important for increasing our lap times. Being able to decelerate the car as quick as possible is crucial for this.

Using simulations we were able to estimate the maximum deceleration possible before the tires lose grip. With this data, we were able to estimate the forces required to bring the car to a stop as fast as possible.

Using this we could calculate the ideal rotor dimensions, calipers and pads that would not have a negative impact on our performance over a long period of time.

Brake biasing is the ratio between the front and rear braking force. With a rearward center of gravity, the rear brakes need to apply more force in the rear in order to lock all wheels at the same time. It also influences the dynamics of the car.

KEEPING IT UPRIGHT



The uprights support the sprung mass of the car through ball bearings at the axles and also serve as a really important component for the stability of the outboard part of the car.

The upright needs to be designed to handle the stresses on it from the bearings as well as the control arms and tie rods.

Unfortunately, a solid block of metal, although structurally

sound, will weigh a lot. Racing is a sport where every gram lost contributes to a better final time and overall integrity of the car.

Therefore, our upright will be designed with an organic shape to save weight.

Based on our final manufacturing limitations, we will optimize the shape for 3D printing or classical CNC milling.

SELECTING THE ECU



The Electronic Control Unit is the brain of the car. This microcontroller controls every single electrical function of the car and is an integral part of any car.

The ECU manages inputs from sensors all over the car and uses the data to optimize other functions such as the motor output or ensuring safety in all the electrical components.

Selecting the right microcontroller for the job required considering more than just the voltage ratings.

Documentation, communication, cost among many other factors were crucial in selecting the right microcontroller for the job.

Each factor is extremely important as it could make or break the car and all of its electrical functions.

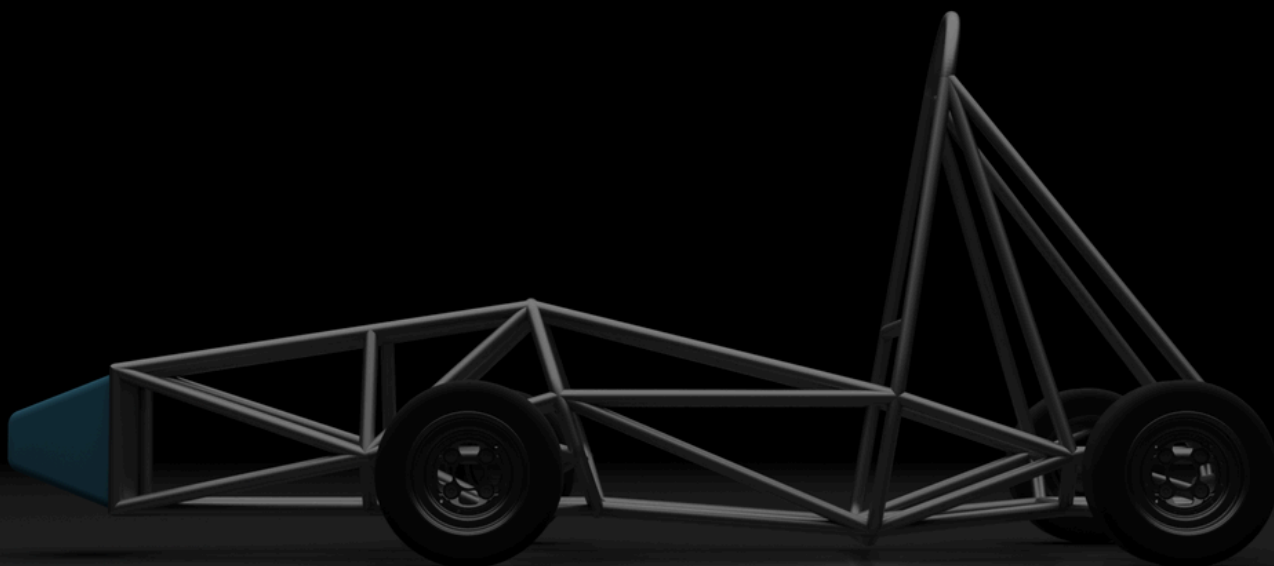
REGISTERING FOR FSUK-CC

We are excited to inform that we have officially registered for formula student UK- concept class!

This is a major milestone for the team as we can finally reap the fruits of our hard work over the last few months. We look forward to representing AUS at an international forum as esteemed as this one and showcasing the talent and effort of our competent engineers.

This will also mark the beginning of our relationship with FS which in the forthcoming years will transition to us participating in FSUK-EV.

FSUK- C has been known to provide the perfect platform for new and transitioning teams to gain feedback on their cost effectively, manufacturing process and design among many other facets.



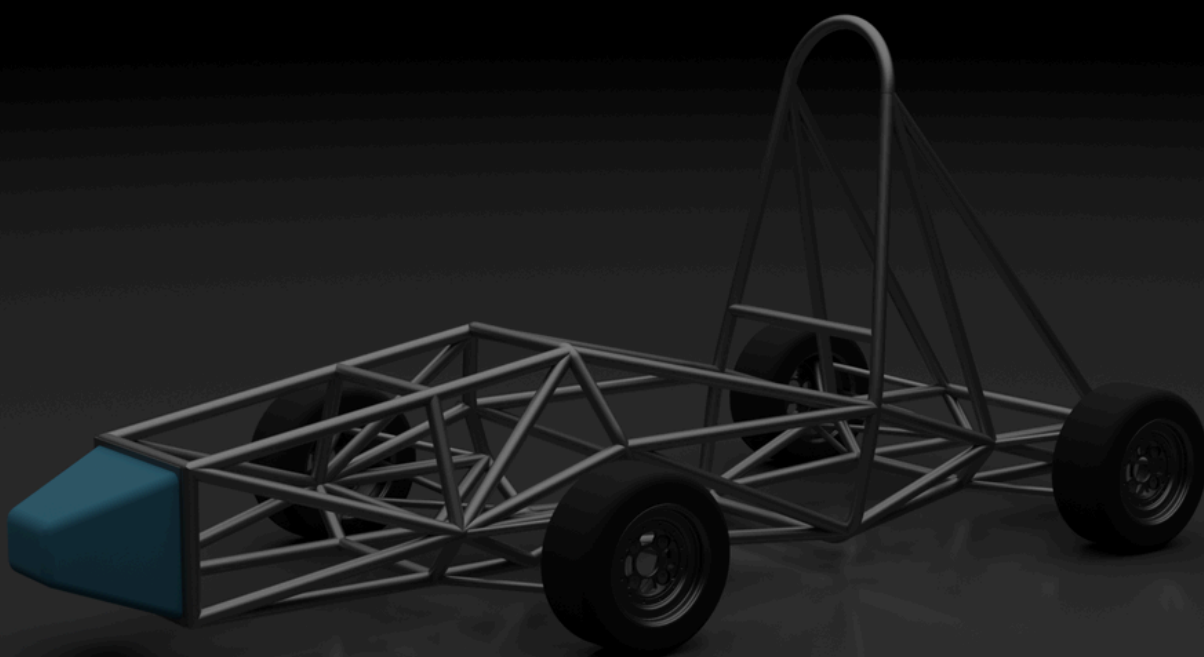
HOW YOU CAN SUPPORT US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your company's answer to any of these questions is yes, contact aaaa@aus.edu to learn more about getting involved with the AUS Racing team.



AUS

American
University
of Sharjah

GOLD

Ansys

BRONZE

AUTOMECH
GROUP